Interaction of high-energy particles: Nuclei (Energy loss processes)

(Lecture 3)

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Outline of this lecture

- Introduction
- Interaction with matter
    - Ionization/excitation, Coulomb collision
    - Nuclear interaction
- Interaction with radiation field
    - e± pair creation
    - Photopion
    - Photo-disintegration
- Radioactive decay
- Interaction with magnetic field
    - Deflection/scattering (will discussed later)

Interaction of high-energy particles: Nuclei

Introduction

- Charged particles undergo electromagnetic interaction when passes through matter.
- Interactions happen mainly with the electrons in the medium.
- Effect of the medium nuclei on the incident particle is negligible.
- Incident nuclei and electrons suffer/behave differently.
  - Electrons suffer both energy losses and heavy scattering from medium electrons.
  - Nuclei suffer mainly energy losses from medium electrons and some small-angle scattering from medium nuclei.

- Nuclei: Keeps almost a straight path while losing energy.
- Electrons: Diffuse easily in the medium while losing energy.

Interaction of high-energy particles: Nuclei

Interaction with matter

Three main processes:

- Ionization/excitation of atoms in neutral medium:

$$X + \text{Atom (neutral)} \longrightarrow \text{Atom}^+ + e^- + X'$$

- Coulomb scattering off the thermal electrons in ionized medium (plasma):

$$X + e^- \longrightarrow e^{-\prime} + X'$$

- Nuclear interactions with nuclei:

$$X + \text{Nuclei (target)} \longrightarrow X' + \text{Nuclei}' + \pi^0 + \pi^\pm \qquad \text{Pion production}$$

$$X\,(A>1) + \text{Nuclei (target)} \longrightarrow X' + \text{Anything} \qquad \text{Disintegration (fragmentation)}$$

- Nuclei suffer energy loss or catastrophic loss due to these processes when passing through matter.
- These interactions can happen inside astrophysical objects and in the interstellar/intergalactic medium.

Interaction of high-energy nuclei with matter

Ionization/excitation in neutral medium

X + Atom (neutral) —> Atom$^+$ + e$^-$ + X'

- Stopping power (energy loss per unit length) of a nuclei of charge *z* and velocity *v* passing through a matter of atomic number density *N* and charge *Z*:

Bethe-Bloch formula: $$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{z^2 e^4 N Z}{4\pi \varepsilon_0^2 m_\mathrm{e} v^2}\left[\ln\left(\frac{2\gamma^2 m_\mathrm{e} v^2}{\bar{I}}\right) - v^2/c^2\right] = z^2 N Z f(v)$$

- Energy loss only depends on *z* and *v* of the incident particle, not on its mass.

Interaction of high-energy nuclei with matter

Ionization/excitation in neutral medium

X + Atom (neutral) —> Atom^{+} + e^{-} + X'

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- Energy loss only depends on z and v of the incident particle, not on its mass.

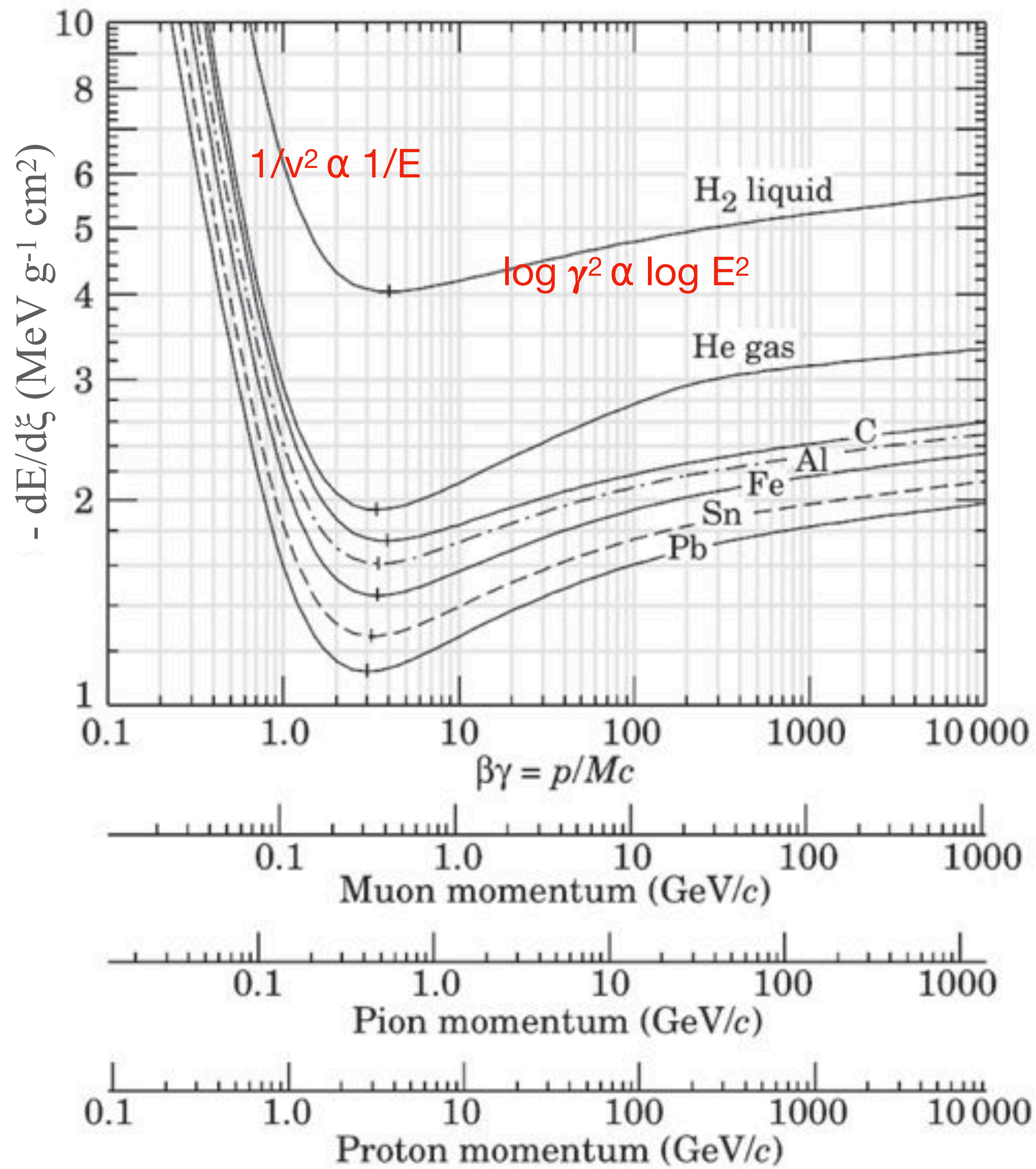
- Mass stopping power: Energy loss in terms of total mass per cross-section of the path.
- If a particle travels a distance x through material of density ρ, then it is said to have traversed ρx kg m^{-2} of the material. Writing $\rho x = \xi$,

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}\xi} = \frac{\mathrm{d}E}{\rho\,\mathrm{d}x}} \Rightarrow -\frac{\mathrm{d}E}{\mathrm{d}\xi} = z^2 f(v)\frac{NZ}{\rho} = z^2 f(v)\frac{Z}{m}$$

where m is the mass of a nucleus of the material. The benefit of expressing the losses in this way is that Z/m is rather insensitive to Z for all the stable elements. For light elements Z/m is $(1/2\,m_{\mathrm{nucl}})$ while for uranium, it decreases to about $(1/2.4\,m_{\mathrm{nucl}})$. Useful in detector construction.

Interaction of high-energy nuclei with matter

Ionization energy loss in different types of material (H_2 liquid, He gas, C, …)



- Important quantity for astrophysical applications is the energy loss rate (dE/dt):

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{\mathrm{d}E}{v\,\mathrm{d}t}$$

Ionization energy loss rate: dE/dt

- Ionization energy loss rate of a nuclei charge of Z and mass M traversing with velocity v through neutral interstellar matter consisting of s sorts of atoms with number density n_s and charge z_s

$$\left(\frac{dE}{dt}\right)_I (\beta \geq \beta_0) = -2\pi r_e^2 c m_e c^2 Z^2 \frac{1}{\beta} \sum_{s=\mathrm{H,He}} n_s [B_s + B'(\alpha_f Z/\beta)]$$

α_f is the fine-structure constant,

$$\beta = v/c \qquad \beta_0 = 1.4e^2/\hbar c = 0.01$$

$$B_s = \left[\ln\left(\frac{2m_e c^2 \beta^2 \gamma^2 Q_{\max}}{\tilde{I}_s^2}\right) - 2\beta^2 - \frac{2C_s}{z_s} - \delta_s\right]$$

γ is the Lorentz factor

$$Q_{\max} \approx \frac{2m_e c^2 \beta^2 \gamma^2}{1 + (2\gamma m_e/M)},$$

$\tilde{I}_{\mathrm{H}} = 19$ eV and $\tilde{I}_{\mathrm{He}} = 44$ eV.

Mannheim & Schlickeiser, 1994, A&A, 286, 983
Strong & Moskalenko, 1998, ApJ, 509, 212

Interaction of high-energy nuclei with matter

Coulomb scattering/collisions in ionized medium (plasma):

X + e$^-$ —> e$^-$' + X'

- Scattering off thermal electrons present in the medium.

Energy loss rate due to coulomb collisions: dE/dt

- The corresponding energy loss rate of a nuclei charge of Z and mass M traveling with velocity $v=\beta c$ through an ionized matter:

$$\left(\frac{dE}{dt}\right)_{\mathrm{Coul}} \approx -4\pi r_e^2\, c m_e c^2 Z^2 n_e \ln\Lambda\, \frac{\beta^2}{x_m^3+\beta^3},$$

r_e is the classical electron radius,
m_e is the electron rest mass,
c is the velocity of light,
n_e is the electron number density in plasma,

$$x_m \equiv [3(\pi)^{1/2}/4]^{1/3}(2kT_e/m_e c^2)^{1/2}, \qquad \ln\Lambda \approx \frac{1}{2}\ln\left(\frac{m_e^2 c^4}{\pi r_e \hbar^2 c^2 n_e}\frac{M\gamma^2\beta^4}{M+2\gamma m_e}\right),$$

T_e is the electron temperature.

Mannheim & Schlickeiser, 1994, A&A, 286, 983
Strong & Moskalenko, 1998, ApJ, 509, 212

Nuclear interactions with matter

- This cannot be considered as a continuous energy loss process.
- A significant fraction (~ 15-20)% of the incident nuclei is lost in a single collision.
- Generally considered as a catastrophic lost in astrophysical calculations.
- That means the incident particle is assumed to get lost after one collision.
- Two channels of interaction: (1) Pion production (mostly for high-energy protons)

  (2) Disintegration/fragmentation (for heavier nuclei)

Nuclear interactions with matter

(1) Pion production (important mostly for protons):

X + Nuclei (target) —> X' + Nuclei' + π^0 + $\pi^\pm$

- Interaction crossection for a high-energy proton interacting with matter proton:

$$\sigma_P(T) = \left(34.3 + 1.88L + 0.25L^2\right)\left[1 - \left(\frac{T_{\rm th}}{T}\right)^4\right]^2 \text{ mb},$$

where T is the total energy of the cosmic ray proton, $L = \ln(T/1\,\text{TeV})$ and $T_{\rm th} = 1.22$ GeV is the threshold energy for the production of π^0 mesons

corresponds to kinetic energy of 282 MeV for the proton

- Interaction time scale, $t_{pp}(E) = 1/(n\, v\, \sigma_p)$, where n is matter density and v is velocity of the incident proton.

- Energy loss rate, $(dE/dt)_{pp} = E/t_{pp}$ where $E = T - m_p c^2$ is the proton kinetic energy.

Kelner, Aharonian, Bugayov, 2006, Phys. Rev. D, 74, 034018

Nuclear interactions with matter

- **Exercise:** Calculate the minimum kinetic energy of a proton required to produce a neutral pion (rest mass 135 MeV) in a proton-proton interaction.

$$p + p \text{ (target) } \longrightarrow p + p + \pi^0$$

[Hint: Start from the invariant property of rest mass]

Ans: About 280 MeV

Nuclear interactions with matter

(2) Disintegration/fragmentation (relevant only for nuclei heavier than proton):

X (A>1) + Nuclei (target) —> X' + Anything

- Interaction crossection for heavy nuclei with matter proton:

$$\sigma_A(E) = \sigma_0 \left[1 - 0.62 e^{-E/0.2} \sin\left(10.9 X^{-0.28}\right)\right],$$

where A represents the mass number of the nuclei, E is the kinetic energy/nucleon in GeV/n, $X = E/0.001$ and

$$\sigma_0 = 45 A^{0.7} \left[1 + 0.016 \sin\left(5.3 - 2.63 \ln A\right)\right] \text{ mb}.$$

- Interaction time scale, $t_{Ap}(E) = 1/(n\, v\, \sigma_A)$

- Energy loss rate, $(dE/dt)_{Ap} = E/t_{Ap}$ where E is the kinetic energy/nucleon.

Letaw J. R., Silberberg R., Tsao C. H., 1983, ApJS, 51, 271

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

- The target photon can be optical starlight, infrared radiation or cosmic microwave background.
- Three main process:

  (1) $e^{\pm}$ pair creation : Relevant for both protons and heavier nuclei.

  (2) Photopion : Relevant mostly for protons in astrophysical environments.

  (3) Photo-disintegration: Relevant only for nuclei heavier than proton.

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

(1) $e^{\pm}$ pair creation: X + photon (target) —> X' + e^+ + e-

Example: p + photon (target) —> p + e^+ + e-

The threshold Lorentz factor of this process is $\boldsymbol{\gamma} \sim m_e/\varepsilon$, where ε is the photon energy. For protons, this corresponds to a threshold energy of ~ $7\mathrm{x}10^{17}$ eV on cosmic microwave background (CMB) of energy 7 x 10^{-4} eV.

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

(1) e± pair creation: X + photon (target) —> X' + e+ + e-

Example: p + photon (target) —> p + e+ + e-

The threshold Lorentz factor of this process is $\boldsymbol{\gamma} \sim m_e/\varepsilon$, where ε is the photon energy. For protons, this corresponds to a threshold energy of ~ 7×10^{17} eV on cosmic microwave background (CMB) of energy 7×10^{-4} eV.

The energy loss rate for a cosmic ray nucleus of charge Ze and Lorentz factor γ is

$$-\left(\frac{dE}{dt}\right)_{e\pm} = \frac{3c\alpha_f\sigma_T}{8\pi} Z^2(m_e c^2)^2 \int_2^\infty dx\, n_\gamma\left(\frac{x m_e c^2}{2\gamma}\right)\frac{\phi(x)}{x^2}$$

where m_e is the electron mass, $\alpha_f = 1/137$ is the fine structure constant, $\sigma_T = 6.65 \cdot 10^{-25}\mathrm{cm}^2$ is the electron Thomson cross section, $\phi(x)$ is an integral over the pair–production cross section and $n_\gamma(\varepsilon)$ is the number density distribution of the target photons. The asymptotic formula for $x \gg 1$,

$$\phi(x \gg 1) \to x[-86.07 + 50.95 \ln x - 14.45(\ln x)^2 + 2.667(\ln x)^3]$$

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

(1) $e^{\pm}$ pair creation: X + photon (target) —> X' + e^+ + e-

Example: p + photon (target) —> p + e^+ + e-

For a background radiation of energy (kT) and energy density u_γ:

$$\left(\frac{dE_\mathrm{p}}{dt}\right)_{\mathrm{e}^\pm} = 2.4 \cdot 10^5 \left(\frac{u_\gamma}{\mathrm{erg\,cm^{-3}}}\right)\left(\frac{\langle kT\rangle}{\mathrm{eV}}\right)^{-2} \frac{\mathrm{eV}}{\mathrm{s}}$$ for protons at the threshold.

Exercise: Calculate dE/dt for the cosmic microwave background radiation (energy, $kT = 7 \times 10^{-4}$ eV and energy density $u_\gamma = 0.26$ eV/cm^3).

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

(2) Photopion production: p + photon (target) —> p + π^0s

—> n + π^+s

The threshold Lorentz factor of this process for the production of ξ pions is:

$$\gamma_{min} \simeq \frac{\xi m_\pi c^2}{2\langle \varepsilon \rangle} \left(1 + \frac{\xi m_\pi}{2 m_p}\right)$$

On CMB, this becomes $\gamma_{min} \simeq \xi \cdot 10^{11}$

This corresponds to a proton threshold energy of ~ 10^{20} eV for one pion production.

This is one possible explanation for the GZK cut-off in the cosmic-ray spectrum.

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction with radiation fields

(2) Photopion production: p + photon (target) —> p + π^0s

—> n + π^+s

For a cosmic ray proton of Lorentz factor γ_p traversing an isotropic photon field of number density $n_\gamma(\varepsilon)$ one obtains the energy loss rate

$$-\left(\frac{dE_p}{dt}\right)_\pi^{(p\gamma)} = \frac{m_p c^3}{2\gamma_p} \cdot \int_{\varepsilon'_{th}/(2\gamma_p)}^{\infty} d\varepsilon n_\gamma(\varepsilon)\varepsilon^{-2} \int_{\varepsilon'_{th}}^{2\gamma_p \varepsilon} d\varepsilon' \varepsilon' \sigma(\varepsilon') \mathrm{K}_p(\varepsilon')$$

where $\sigma(\varepsilon')$ and $\mathrm{K}_p(\varepsilon')$ are the total photo–hadron production cross section and inelasticity, respectively, as a function of the photon energy in the proton rest frame. The proton rest frame threshold energy ε'_{th} is given by

$$\varepsilon'_{th,\xi\pi} = \xi m_\pi c^2 \left[1 + \frac{\xi m_\pi}{2m_p}\right]$$

for the production of ξ pions so that for the production of a single pion the rest–system threshold energy is $\varepsilon'_{th,\pi} = 145$ MeV.

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction of high-energy particles: Nuclei

Interaction with radiation fields

(2) Photopion production: p + photon (target) —> p + π^0s

—> n + π^+s

For a background radiation of energy (kT) and energy density u_γ:

$$\left(\frac{dE_\mathrm{p}}{dt}\right)_\pi^{(\mathrm{p}\gamma)} = 1.8 \cdot 10^{10} \left(\frac{u_\gamma}{\mathrm{erg\,cm^{-3}}}\right)\left(\frac{\langle kT\rangle}{\mathrm{eV}}\right)^{-2} \frac{\mathrm{eV}}{\mathrm{s}}$$

Exercise: Calculate dE/dt for the cosmic microwave background radiation (energy, kT =7 x 10^{-4} eV and energy density u_γ=0.26 eV/cm^3).

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction with radiation fields

(3) Photo-disintegration: X + photon (target) —> (X-1) + p,n

- This process is important only for nuclei heavier than protons.

The photodisintegration rate for a nucleus of mass A with the subsequent release of i nucleons in an isotropic photon field of number density $n_\gamma(\varepsilon)$ is given by the expression

$$R_{A,i} = \frac{c}{2}\gamma^{-2}\int_0^\infty d\varepsilon n_\gamma(\varepsilon)\varepsilon^{-2}\int_0^{2\gamma\varepsilon} d\varepsilon'\varepsilon'\sigma_{A,i}(\varepsilon') \qquad (3.10),$$

where γ is the Lorentz factor of the cosmic ray nucleus and ε' the photon energy measured in the rest frame of the nucleus.

Stecker, 1969, Phys. Rev. 180, 1264

Interaction of high-energy particles: Nuclei

Radioactive decay

- Only important for unstable nuclei such as beryllium 10 (Be^{10}), aluminum 26 (Al^{26}) …
- In high-energy astrophysics, most of these isotopes are produced by the fragmentation of heavier nuclei. E.g: Be^{10} is produced from carbon and oxygen.
- Flux ratio of radioactive to stable isotopes produced from the same interaction gives a direct measure of the cosmic-ray confinement time in the Galaxy.

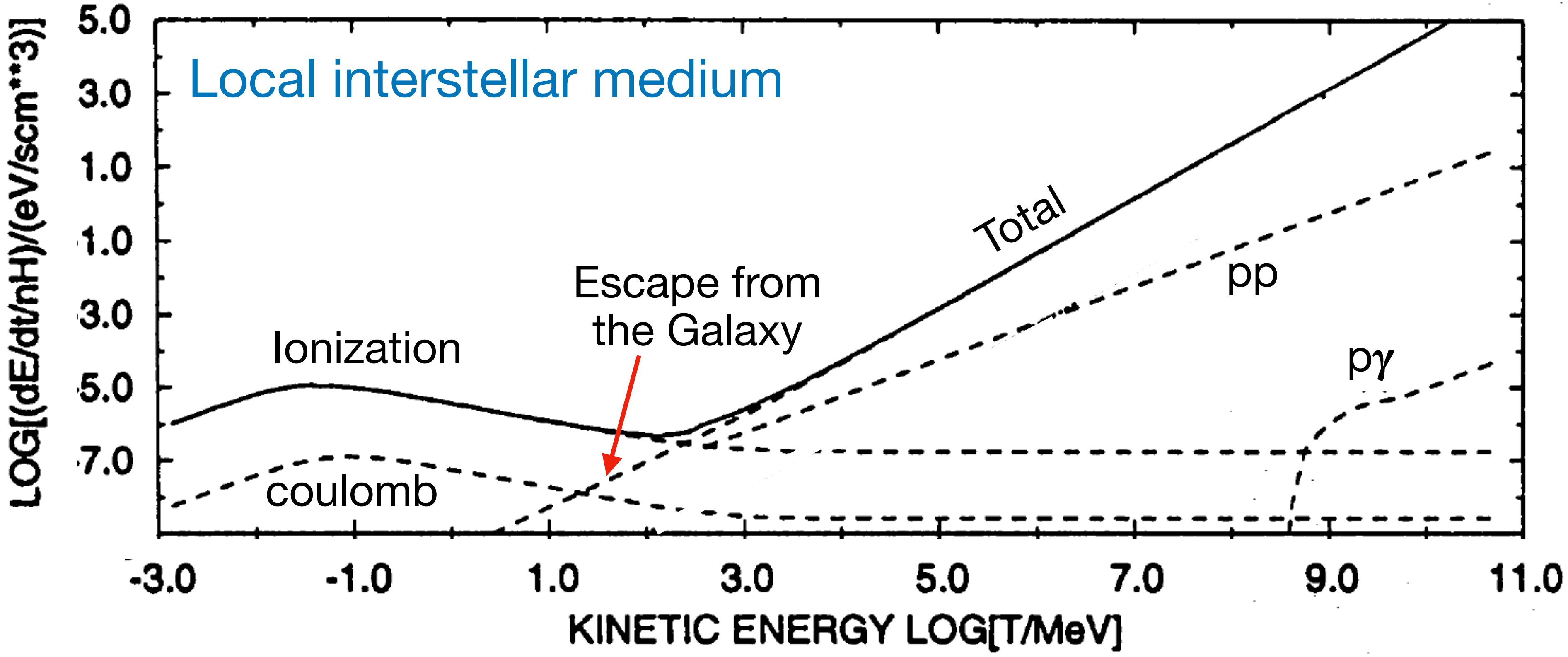
Table 15.5 Decay half-lives of some important radioactive isotopes created in spallation reactions (Yanasak *et al.*, 2001).

Isotope	Decay mode	Half-life of decay mode
^{10}Be	β^-	1.51×10^6 years
^{14}C	β^-	5700 years
^{26}Al	β^+	8.73×10^5 years
	e capture	8.45×10^6 years
^{36}Cl	β^-	3.07×10^5 years
	e capture	1.59×10^7 years
^{54}Mn	β^-	$(6.3 \pm 1.7) \times 10^5$ years
	e capture	312 days

High Energy Astrophysics, Longair

Interaction of high-energy particles: Nuclei

Total energy loss rate for cosmic-ray protons in the interstellar medium



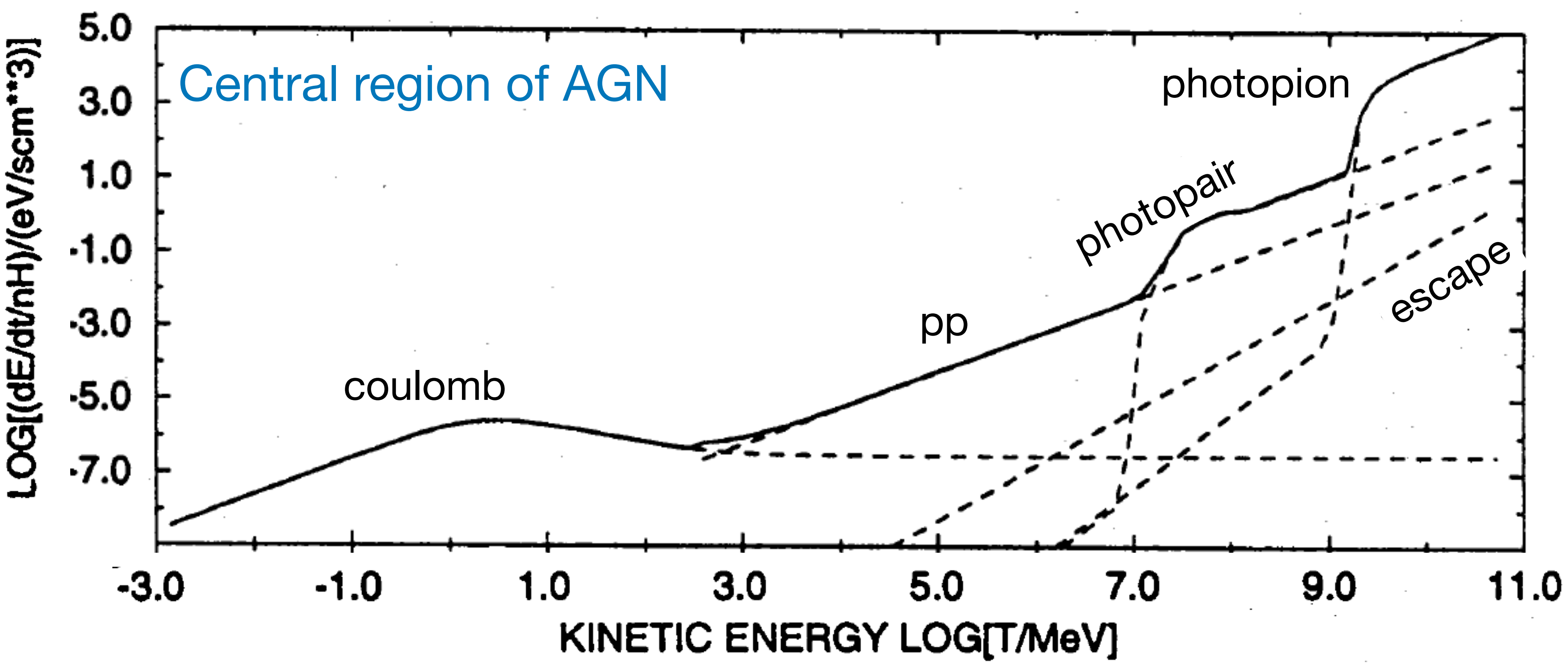
Parameter values used in the calculation:
- Neutral gas density = 1.14 cm^{-3}
- Thermal electron density = 0.01 cm^{-3}
- Temperature of the plasma = 3×10^5 K
- Luminosity of thermal radiation for $p\gamma$ interaction = 10^{44} erg/s

  and photon energy = 1 eV

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Interaction of high-energy particles: Nuclei

Total energy loss rate for protons in the central region of Active galactic nuclei



Parameter values used in the calculation:

- Thermal electron/matter density = 3 x 10^5 cm^{-3}
- Temperature of the plasma = 10^7 K
- Radius of the compact region = 10^{-3} pc
- Luminosity of thermal radiation for pγ interaction = 10^{44} erg/s and photon energy = 30 eV
- Equipartition assumed between magnetic field density and radiation.

Mannheim & Schlickeiser, 1994, A&A, 286, 983

Next, we will learn about the interaction of high-energy electrons.